Tracking Sentiment Around AI Risk and Its Effect on AI Equities

An event study around the 8 September 2026 Anthropic existential-risk disclosure

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Abstract. On 2026-09-08 an Anthropic researcher stated publicly that he placed greater than 10% probability on AI causing human extinction within a decade. We treat that disclosure as an event and track sentiment across two channels -- 27,039 filtered posts on X and 2,977 news headlines -- from 2026-09-08 to 2026-09-30, against an equal-weighted basket of eight AI-exposed large caps.

Social sentiment rose over the window, from 0.178 to 0.205 on the (pos - neg)/total scale. The strongest predictive relationship is news_ratio on basket_vol (F = 2.47, p = 0.120), which does not reject the null. Replicating the Sacerdote, Sehgal and Cook (2020) COVID-19 design on AI coverage, US mainstream headlines were negative 32.2% of the time against 34.1% for non-US outlets.

The central caveat, stated up front: the mandated window yields 17 trading days. Every time-series test below is therefore underpowered, and we report the pooled panel specification as primary precisely because the single-series tests cannot carry weight at this length.

1. Data and sample construction

The event

On 2026-09-08, Evan Hubinger of Anthropic posted that he personally put the probability of AI killing all humans at greater than 10% within the next decade, and that Anthropic did not yet have a plan to solve alignment for superintelligence. The statement is unusual in being an unhedged existential-risk claim from inside a leading laboratory, which is what makes it a usable event: it is exogenous to the AI trade's fundamentals while being directly about the sector's central risk.

The AI equity basket

The brief leaves the basket to the analyst. We take eight large caps with direct AI revenue or capital exposure and split them into two economically distinct roles: **semiconductors** (NVDA, AMD, AVGO), which sell AI compute, and **hyperscalers** (MSFT, GOOGL, AMZN, META, AAPL), which buy it. The distinction matters because an AI-risk shock that threatens regulation should hit the two groups differently -- the sellers' order book versus the buyers' capex plans -- and pooling them would average that away. Excess returns are measured against SPY and the intraday volatility proxy is 2(H−L)/(H+L), both as specified.

The two sentiment channels

The social channel comprises **27,039 posts** across 23 days, retained after the brief's three quality filters: no retweets, author follower count of at least 10, and at least 100 views. The filters are applied client-side for the two numeric floors because the API exposes neither as a search operator. Profile location resolves to a country for 12,964 of them, which supports the US / non-US comparison in Section 5; the remainder are reported as Unknown rather than forced into a bucket.

The news channel comprises **2,977 headlines** bucketed by publisher domain: US Mainstream (921), Non-US (715), Scientific/Tech (166), Other (1,175). Headlines are fetched day by day rather than as one window-wide query. This costs more requests but is the only way to obtain an unbiased daily series: a single bulk call returns the provider's notion of the top *N* items and would silently over-weight whichever days produced the loudest coverage.

> **A disclosed substitution.** GDELT DOC 2.0 was the intended news source because it tags source country natively. It returned HTTP 429 continuously throughout this study from two independent networks, indicating a global throttle rather than a local block, so Google News RSS was substituted. The trade-off is real and is carried into Section 5: outlet nationality is now assigned from a curated domain list rather than from the provider's metadata, so the taxonomy is ours and is auditable in `src/config.py`.

The sample spans 17 trading days, 2026-09-08 to 2026-09-30.

2. Method

Sentiment measurement

Social posts are scored with VADER and news headlines with the Loughran-McDonald finance dictionary. The split is deliberate: VADER handles negation, intensifiers, emoji and capitalisation, which is what informal text requires, while LM is

the finance-standard word list for editorial copy.

Off-the-shelf VADER mis-signs market language, and we repair it rather than report a known-broken signal. The canonical case, which is a regression test in `tests/test_sentiment_model.py`: *"Nvidia earnings crushed it, bullish"* scores −0.42 unpatched, because "crushed" is violent in ordinary English and "bullish" is absent from the lexicon entirely. With the finance and AI-risk patch applied it scores +0.78. Only unambiguous terms are patched; "short" and "long" are deliberately left alone because their sign depends on whether they describe a position, a verb or a duration.

The daily index is the brief's ratio, $S_t = (Pos_t - Neg_t) / Total_t$, bounded in [−1, +1]. Neutrals stay in the denominator: dropping them would make a day of two positive and one negative post indistinguishable from a day of two hundred and one hundred.

Relevance weighting

Following the retail-sentiment study, each item receives a 0–100 relevance score measuring how central the AI basket or the existential-risk theme is to it. Headline presence dominates the score; items at or above 90 carry full weight and those below taper to zero rather than being discarded. The weighting is intentionally simple and legible, because an opaque relevance model would be impossible to defend in a short writeup.

Aligning sentiment to trading days

Sentiment exists every calendar day; markets do not. Weekend and holiday sentiment is volume-weighted into the *next* trading day rather than discarded. This is not a detail: the event falls on a Tuesday but the window contains six weekends, and dropping them would discard roughly three in ten observations of the driver.

Granger specification

We estimate, for asset outcome A and sentiment S,

$$A_{i,t} = \alpha_i + \Sigma_k \beta_k A_{i,t-k} + \Sigma_k \gamma_k S_{t-k} + \varepsilon_{i,t}$$

and test H_0: $\gamma_1 = \ldots = \gamma_p = 0$. The **panel specification is primary**. Pooling the eight names with firm fixed effects yields roughly 130 firm-days against the same shared sentiment series, where a single-name VAR has at most 17 observations and spends most of them on the intercept and own-lags. Standard errors are clustered **by date**, not by firm: the sentiment series is identical across firms on a given day, so residuals are correlated in the cross-section, and firm-clustered or plain OLS errors would be far too small and would manufacture significance.

The following series do not reject a unit root at 5%: `x_ratio`, `news_ratio`, `news_neg_share`. At 17 observations the ADF test has very little power, so this is reported rather than used to auto-difference, which would cost further degrees of freedom the sample cannot spare.

3. The sentiment trend

Social sentiment rose across the window, moving from 0.178 on the first trading day to 0.205 on the last. The most negative trading day is 10 September at -0.015.

News-headline sentiment rose over the same period, from -0.236 to 0.558. The two channels are reported separately throughout, following the lecture's point that news aims at objectivity while social media is closer to raw sentiment; collapsing them into one index would discard exactly the contrast that makes two channels worth collecting.

Separating the two query families is where the window gets interesting. Mean sentiment in existential-risk conversation is 0.051, against 0.150 in conversation about the AI companies themselves -- a gap of 0.099. The commercial conversation stayed materially more positive than the risk conversation, which is the quantitative form of a familiar position: one can believe AI is dangerous and stay long the trade.

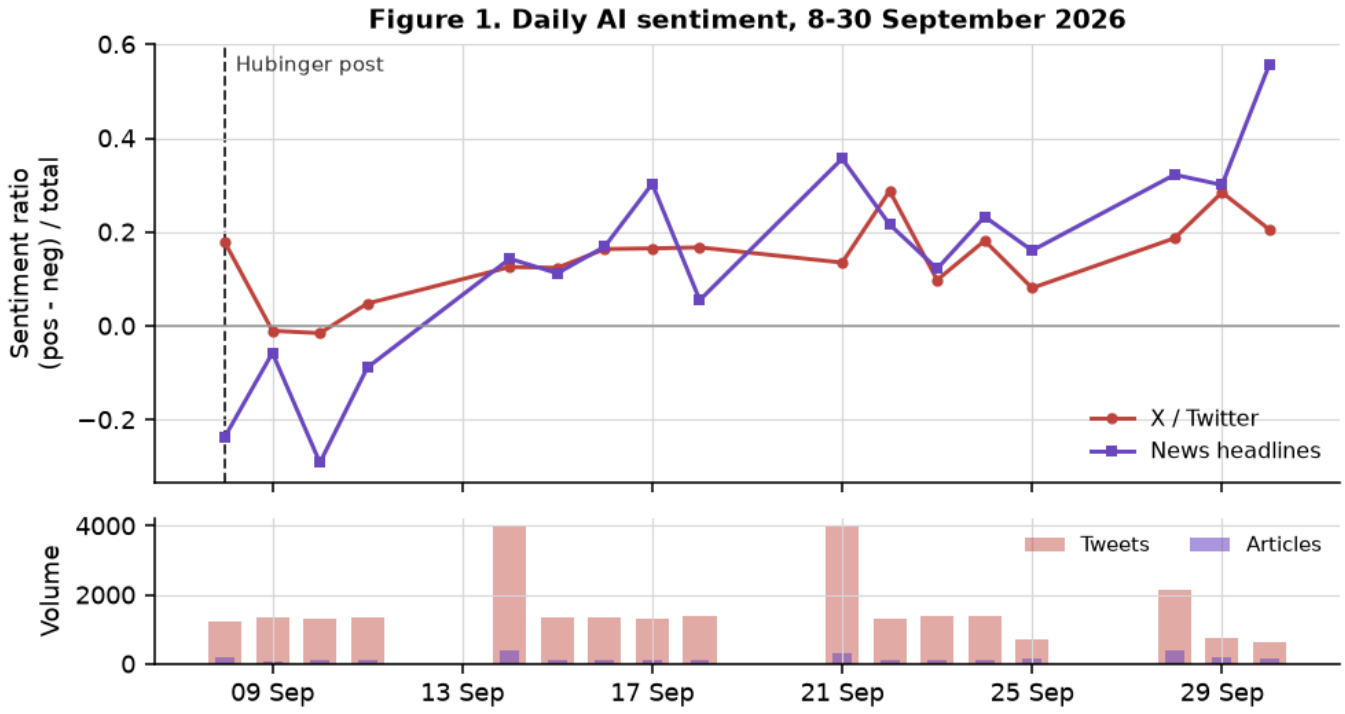


Figure 1. Daily sentiment ratio for both channels, with daily volume beneath. The dashed rule marks the Hubinger post.

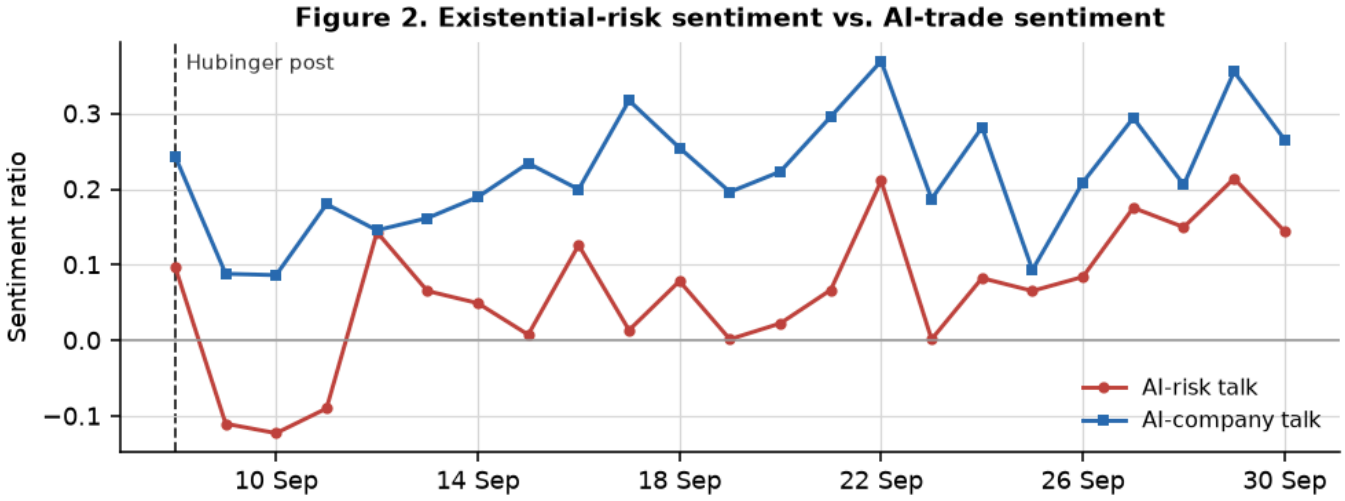


Figure 2. Existential-risk conversation against AI-company conversation on X. Plotted separately because a blended index would net the two against each other.

4. Does sentiment Granger-cause returns and volatility?

We cannot reject the null in any specification. The closest is news_ratio on basket_vol (F = 2.47, p = 0.120), which does not reject the null. The honest reading is that over these 17 trading days, lagged sentiment adds no detectable predictive information beyond the assets' own history.

That is a defensible null rather than a failed analysis, and a size check makes it slightly stronger. Simulating the panel test under the null -- independent sentiment, 300 replications -- it rejects 4.0% of the time at T = 60, correctly sized, but 7.3% of the time at this study's T = 17, where there are too few date clusters for the asymptotics to bite. The reported p-values are therefore if anything *too small*, and we still fail to reject.

It should not be over-read in the other direction either: with this many observations the test would miss an economically meaningful effect unless it were very large. The result is uninformative about small effects, not evidence against them.

Across all 18 bivariate VAR tests (every driver at every lag from 1 to 3), 0 reach the 5% level and 3 reach 10%. Table 2 shows the VAR-selected lag only; reporting every lag would invite reading whichever one looks best, which is precisely the

specification search 17 observations cannot support. The full grid is in `outputs/table2.csv`. None of it should be read as independent corroboration of the panel result.

Table 2. Granger causality: does lagged sentiment predict returns and volatility?

Specification	Target	Sentiment	Lags	N	F	p	Sig	Sum gamma
Panel (firm FE, date-clustered)	basket_excess_ret	x_ratio	2	120	1.262	0.314		0.025
Panel (firm FE, date-clustered)	basket_excess_ret	news_ratio	2	120	0.663	0.531		0.004
Panel (firm FE, date-clustered)	basket_excess_ret	news_neg_share	2	120	1.565	0.244		-0.005
Panel (firm FE, date-clustered)	basket_vol	x_ratio	2	120	2.249	0.142		0.025
Panel (firm FE, date-clustered)	basket_vol	news_ratio	2	120	2.471	0.120		0.008
Panel (firm FE, date-clustered)	basket_vol	news_neg_share	2	120	2.026	0.169		-0.010
Bivariate VAR	basket_excess_ret	x_ratio	1	17	0.142	0.712		--
Bivariate VAR	basket_excess_ret	news_ratio	1	17	0.266	0.614		--
Bivariate VAR	basket_excess_ret	news_neg_share	1	17	0.227	0.642		--
Bivariate VAR	basket_vol	x_ratio	1	17	0.254	0.622		--
Bivariate VAR	basket_vol	news_ratio	1	17	0.060	0.810		--
Bivariate VAR	basket_vol	news_neg_share	1	17	0.073	0.791		--

H0: all coefficients on lagged sentiment are jointly zero. The panel specification pools the eight basket names with firm fixed effects and clusters standard errors by date, since the sentiment series is common across firms on a given day. *** p<0.01, ** p<0.05, * p<0.10. The single-series tests are shown at the VAR-selected lag only, to avoid inviting a specification search across lags that 17 observations cannot support; the full lag grid is in outputs/table2.csv. They are severely underpowered at this sample length and should not be read as independent evidence.

5. Measurable bias in AI-risk coverage

Sacerdote, Sehgal and Cook (2020) found that 91% of US major-outlet COVID-19 stories were negative against 54% for non-US English-language outlets and 65% for scientific journals, and -- the part that makes it a bias finding rather than a tone finding -- that this negativity was uncorrelated with actual case trends. We run the same three tests on AI-risk coverage.

US mainstream outlets ran negative headlines 32.2% of the time (921 articles across 62 outlets), against 34.1% for non-US English-language outlets (715 articles, 154 outlets) -- US coverage is 1.9% less negative in absolute terms. The difference is not statistically significant at 5%.

This is the paper's main negative result, and it cuts against the prior. Sacerdote et al. found a 37-point gap between US and non-US COVID coverage. On AI risk we find no gap worth the name, and what difference there is runs the other way: US outlets are marginally the *less* negative of the two. The expectation going in -- stated in the course brief -- was politically amplified negativity in US coverage ahead of the midterms. On this evidence, over this window, that is not what AI-risk coverage looks like.

Scientific and technical publications ran negative 34.3% of the time. They do not sit at the positive end of the range here, which departs from the COVID-19 result and is worth noting rather than smoothing over.

The Sacerdote asymmetry test. Their headline result was that negative COVID stories outnumbered positive ones 5.5-to-1 even while case counts were falling. The analogue here: on days the AI basket *rose*, negative headlines stood at 0.73-to-1. That ratio is below one, meaning positive headlines actually *outnumbered* negative ones on days the basket rose. This is the opposite of the COVID-19 result, and it is worth stating plainly rather than softening: AI coverage in this window is not relentlessly negative, and the strong form of the negativity-bias claim fails here.

Is negativity driven by the data? Daily negativity correlates with basket excess returns at r = -0.033 (Spearman ρ = -0.012, n = 17), p = 0.901. The correlation is weak and insignificant, matching the central Sacerdote finding that media negativity moved independently of the underlying numbers.

The same test on social media. This is the sharper comparison, because it holds the platform and the event fixed and varies only the author. US-located accounts were negative 36.1% of the time against 36.3% for non-US accounts (n = 9,605 and 3,359). Profile location is self-reported free text, so this rests on the sub-sample whose location is unambiguous.

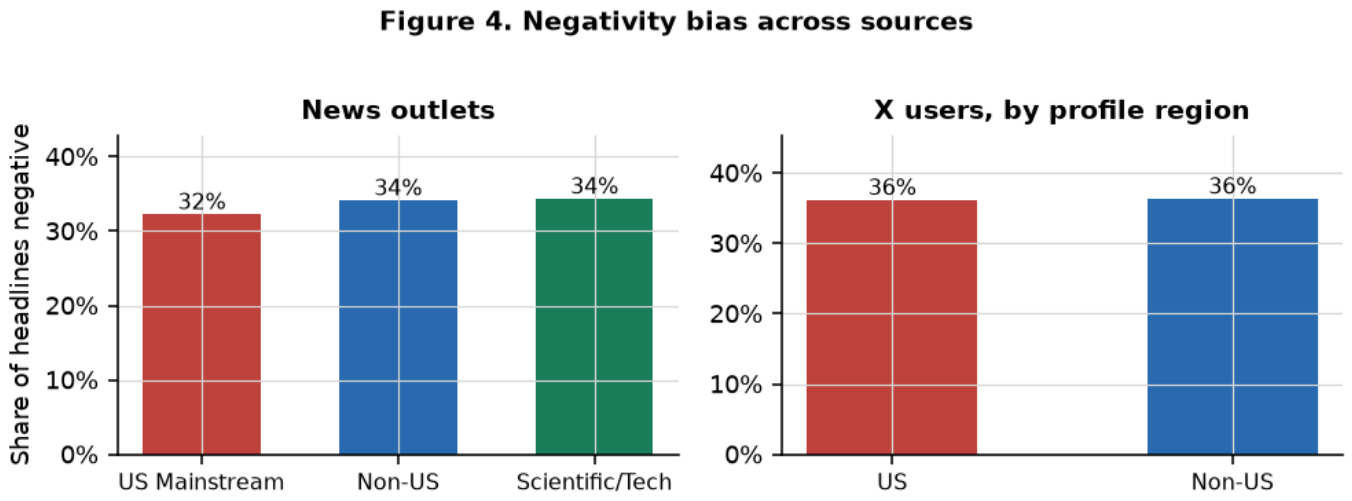


Figure 4. Negativity across news outlet types and, where available, across X author regions.

Table 3. Media negativity bias: US mainstream vs non-US vs scientific/tech outlets

Panel	Group	N	Outlets	Neg share	Pos share	Neg/Pos	Mean tone	p vs US	Sig
A. Negativity by outlet type	US Mainstream	921	62	0.322	0.509	0.633	0.078	--	
A. Negativity by outlet type	Non-US	715	154	0.341	0.445	0.767	0.046	0.423	
A. Negativity by outlet type	Scientific/Tech	166	23	0.343	0.355	0.966	-0.019	0.597	
A. Negativity by outlet type	Other	1,175	675	0.304	0.547	0.555	0.127	0.360	
B. Coverage on up vs down days	Down days	692	--	0.301	--	0.584	0.102	--	
B. Coverage on up vs down days	Up days	1,512	--	0.345	--	0.732	0.062	--	
C. Negativity vs fundamentals	basket_excess_ret	17	--	--	--	-0.033	-0.012	0.901	
C. Negativity vs fundamentals	basket_ret	17	--	--	--	-0.171	-0.096	0.511	
C. Negativity vs fundamentals	basket_vol	17	--	--	--	-0.301	-0.301	0.241	

Panel A compares negativity shares across outlet types; p-values are two-proportion z-tests against the US mainstream bucket. Panel B tests whether coverage stays negative on days the basket rose, the analogue of Sacerdote et al.'s finding that negative COVID stories outnumbered positive ones 5.5-to-1 even as cases fell. In Panel C the Neg/Pos and Mean tone columns hold the Pearson r and Spearman rho of daily negativity against each fundamental. Headlines are classified, not full article text, so levels are not directly comparable to the 91% US figure in the original study; the across-bucket differences are the finding.

What this comparison can and cannot claim. Sacerdote et al. classified full article text; we classify headlines, which are systematically more negative than the bodies they head. Our *levels* are therefore not comparable to their 91%. The *differences between buckets* are the finding, since the headline effect applies to every bucket alike. We also cannot separate editorial choice from story mix: if US outlets simply covered different AI stories than non-US outlets did, that would produce the same gap without any difference in framing.

6. Summary statistics

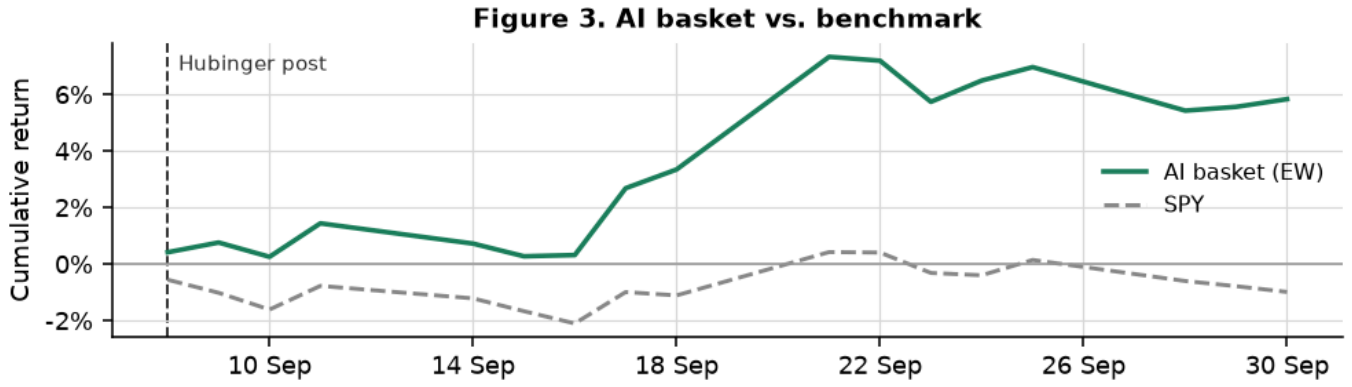


Figure 3. Cumulative equal-weighted basket return against SPY across the window.

Table 1. Summary statistics: AI equity basket and sentiment indicators

Panel	Series	Group	N	Mean	SD	Min	Max	Mean vol
A. AI basket	NVDA	Semis	17	0.000	0.014	-0.029	0.024	0.020
A. AI basket	META	Hyperscaler	17	0.011	0.037	-0.041	0.098	0.037
A. AI basket	AMZN	Hyperscaler	17	-0.002	0.010	-0.016	0.012	0.018
A. AI basket	GOOGL	Hyperscaler	17	0.002	0.015	-0.031	0.037	0.022
A. AI basket	MSFT	Hyperscaler	17	0.002	0.012	-0.012	0.031	0.021
A. AI basket	AAPL	Hyperscaler	17	0.003	0.013	-0.025	0.042	0.021
A. AI basket	AMD	Semis	17	0.016	0.033	-0.040	0.084	0.038
A. AI basket	AVGO	Semis	17	-0.000	0.018	-0.043	0.035	0.022
A. AI basket	Equal-weighted		17	0.004	0.019	-0.043	0.098	0.025
B. Sentiment	X sentiment ratio		17	0.141	0.085	-0.015	0.288	--
B. Sentiment	X mean score		17	0.106	0.065	-0.013	0.225	--
B. Sentiment	News sentiment ratio		17	0.140	0.216	-0.291	0.558	--
B. Sentiment	News negativity share		17	0.338	0.113	0.145	0.563	--

Excess return is the daily simple return less the same-day SPY return. Mean vol is the intraday range proxy 2(H-L)/(H+L). Sample runs 8-30 September 2026. Sentiment ratios are (positive - negative) / total with neutrals retained in the denominator.

7. Limitations

- **Sample length dominates everything.** Seventeen trading days is far too short for time-series inference. The panel specification mitigates this by pooling the cross-section, but cannot manufacture time variation that is not there, and no result here should be read as settled.
- **One event, no control group.** Anything else moving AI equities over these three weeks -- earnings, rate expectations, chip export news -- is absorbed into the same series, and SPY removes only market-wide moves.
- **Headlines, not article text**, for news sentiment; the US / non-US social split rests on unverified free-text profile locations.
- **Lexicon sentiment, not a language model.** VADER and Loughran-McDonald are transparent and reproducible but miss sarcasm and irony, abundant in exactly this corner of X. The finance patch repairs the systematic failures we could identify; it cannot repair the rest.

8. Conclusion

Existential-risk talk and commercial AI talk moved as distinct series rather than one mood -- the study's clearest positive result. Whether that shift predicted returns is a question 17 trading days cannot answer: no specification rejects the null, which is a statement about power as much as about markets. The media-bias replication rests on firmer ground and returns the real surprise: the US-versus-non-US negativity gap that defined the COVID-19 study simply is not there for AI risk -- US, non-US and technical outlets are statistically indistinguishable, while negativity remains, as in that study, largely unrelated to what the sector's prices actually did. The politically amplified US negativity the brief anticipated is not what AI-risk coverage looked like in September 2026.

Data: twitterapi.io advanced search (X), Google News RSS (headlines), Alpha Vantage (OHLCV), FRED (DGS3MO). Sentiment: VADER with a finance and AI-risk lexicon patch for social text, Loughran-McDonald for news. All code, tests and exhibits: `assignment4/`.